

Week 4, Day 2 : Preprocessing & Model Comparison

Preprocessing

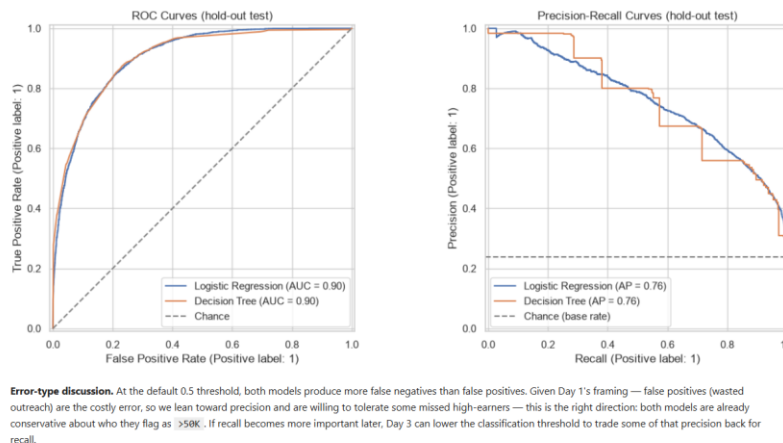
Numeric features (age, education-num, capital-gain, capital-loss, hours-per-week): median imputation -> StandardScaler. Median chosen over mean for robustness to skew/outliers in capital-gain/loss. flwgt (a Census sampling weight, not a person-level feature) was dropped; removing it changed all metrics by <0.001, empirically confirming it carried no signal. Categorical features: most-frequent imputation -> one-hot encoding (handle_unknown='ignore'). A log1p transform on capital-gain/loss was tested to address skew but was reverted because it measurably worsened Logistic Regression (Precision 0.7264 to 0.7045, F1 0.6576 to 0.6417, ROC AUC 0.9040 to 0.8960) while leaving the Decision Tree unchanged, since the feature's signal appears to be a threshold effect (any gain vs. none) that log-compression blurs rather than clarifies.

Model Comparison (Hold-Out Test Set)

Model	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
Majority-class	0.7607	0.0000	0.0000	0.0000	0.5000	0.2393
Rule-based (edu-num $\geq$ 13 OR cap-gain>0)	0.7452	0.4746	0.6031	0.5312	0.7426	0.4946
Logistic Regression	0.8501	0.7264	0.5997	0.6570	0.9038	0.7598
Decision Tree	0.8578	0.7938	0.5483	0.6486	0.9048	0.7597

Both models comfortably outperform the rule-based baseline. Logistic Regression and Decision Tree both improve precision well beyond the baseline (0.4746), while maintaining ROC AUC $\approx$ 0.90 and PR AUC $\approx$ 0.76. Decision Tree achieves the highest precision (0.7938), whereas Logistic Regression provides slightly higher recall and a more interpretable decision boundary.

ROC & Precision-Recall Curves



Both models trace nearly identical ROC and Precision-Recall curves, indicating very similar ranking performance (ROC AUC $\approx$ 0.90, PR AUC $\approx$ 0.76). Despite this, **Decision Tree achieves noticeably higher precision (0.7938 vs. 0.7264)** at the default threshold, making it the stronger model under this project's precision-first objective. The Decision Tree's PR curve appears stepped because it predicts a finite number of probability values (104 leaves), whereas Logistic Regression produces smooth continuous probabilities.

Model Selection & Next Steps

Continuing with both models into Day 3. Logistic Regression is the interpretable, production-friendly candidate because its coefficients (capital-gain, marital status, education-num as top drivers) directly validate Day 1's manual error analysis. The Decision Tree captures nonlinear/threshold interactions a linear model cannot (e.g., splitting on capital-gain twice within one branch), while its 0.0025 train/test accuracy gap confirms it is not overfitting at depth 8. Since both models achieve nearly identical ROC AUC and PR AUC, **precision becomes the deciding factor**. Decision Tree achieves the highest precision (0.7938), making it the strongest model under this project's primary metric, whereas Logistic Regression provides slightly higher recall and a more interpretable decision boundary.

Planned for Day 3: engineer a has_capital_gain feature (capital-gain > 0), evaluate class weighting, regularization and depth tuning, test Histogram Gradient Boosting as a stronger nonlinear model, and investigate higher decision thresholds to further improve precision where appropriate.